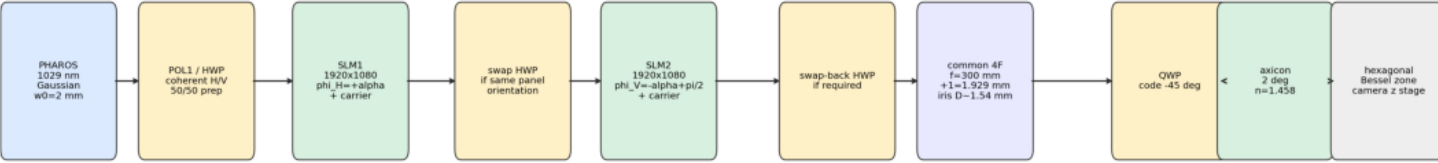


architecture strip was unreadable (~3-4 pt) and clipped; rebuilt as a two-row wrapped vector diagram

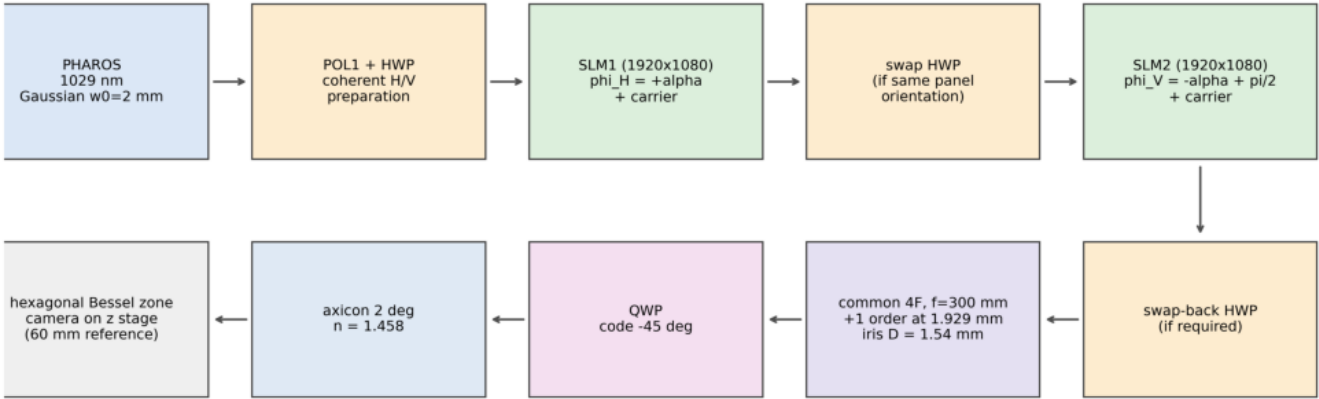
BEFORE: F08

MODE 2W-FIX Figure 1: sequential single-beam architecture (no PBS split, no H/V interferometer arms)



Alternate valid branch: if SLM2 is mounted with orthogonal LC director, swap/swap-back HWPs may be omitted after the panel-orientation test.

AFTER: F1_sequential_architecture.png
Sequential single-beam dual-SLM architecture (canonical)

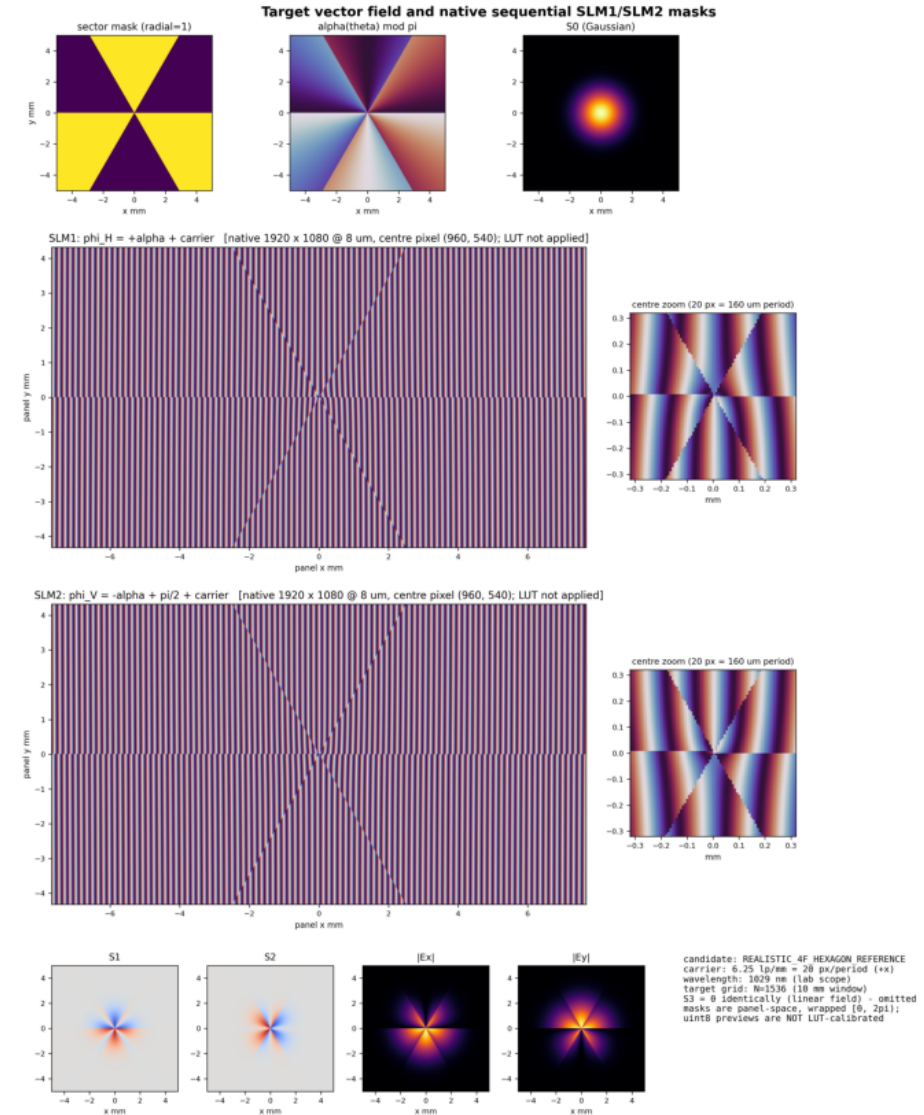
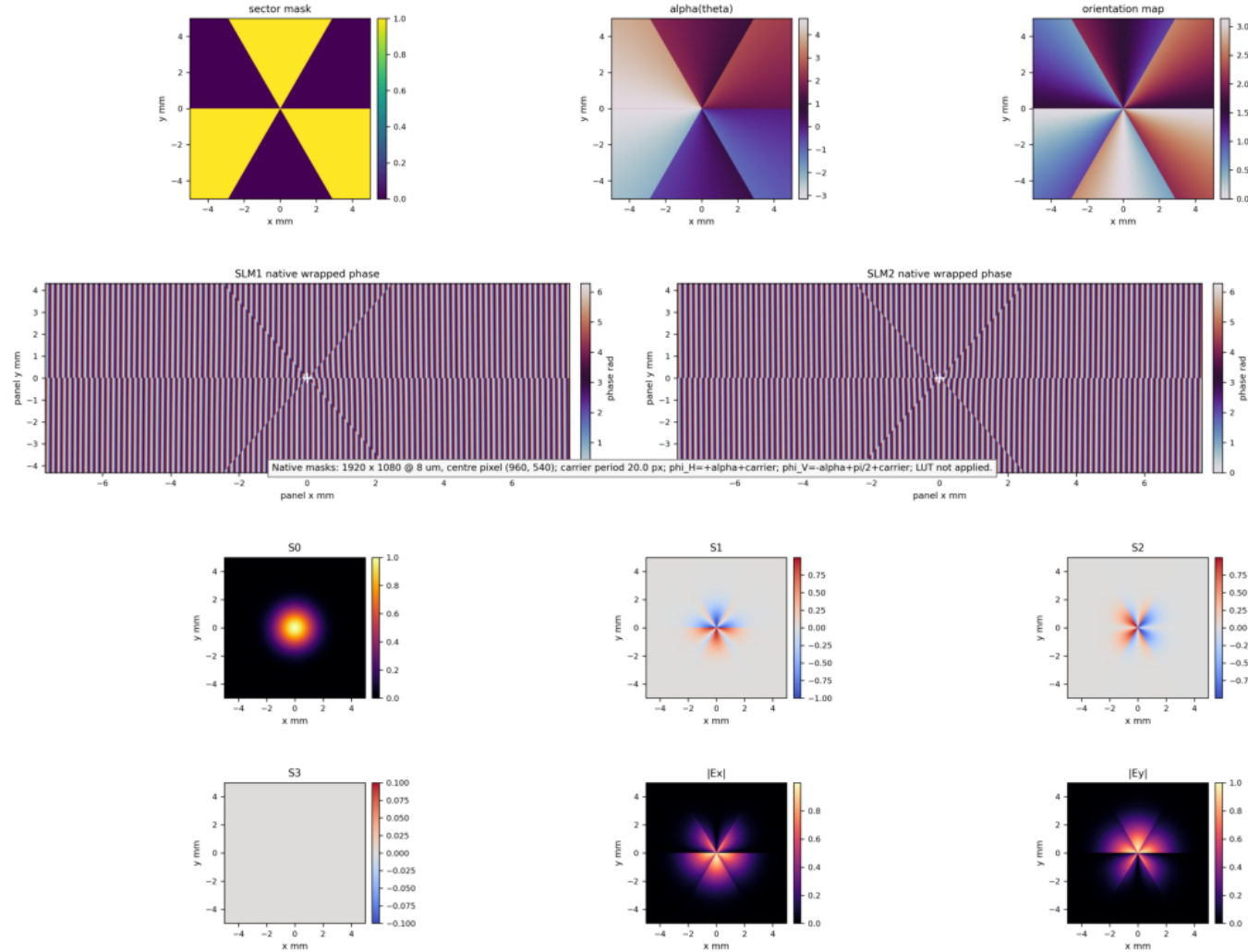


Sequential single-beam architecture (no PBS split, no H/V interferometer arms). Alternate valid branch: mount SLM2 with orthogonal LC director and omit both swap HWPs.

native masks were too small to inspect; rebuilt with full-width panels, carrier zoom inset and no dead S3 panel
 BEFORE: F09

AFTER: F2_target_and_masks.png

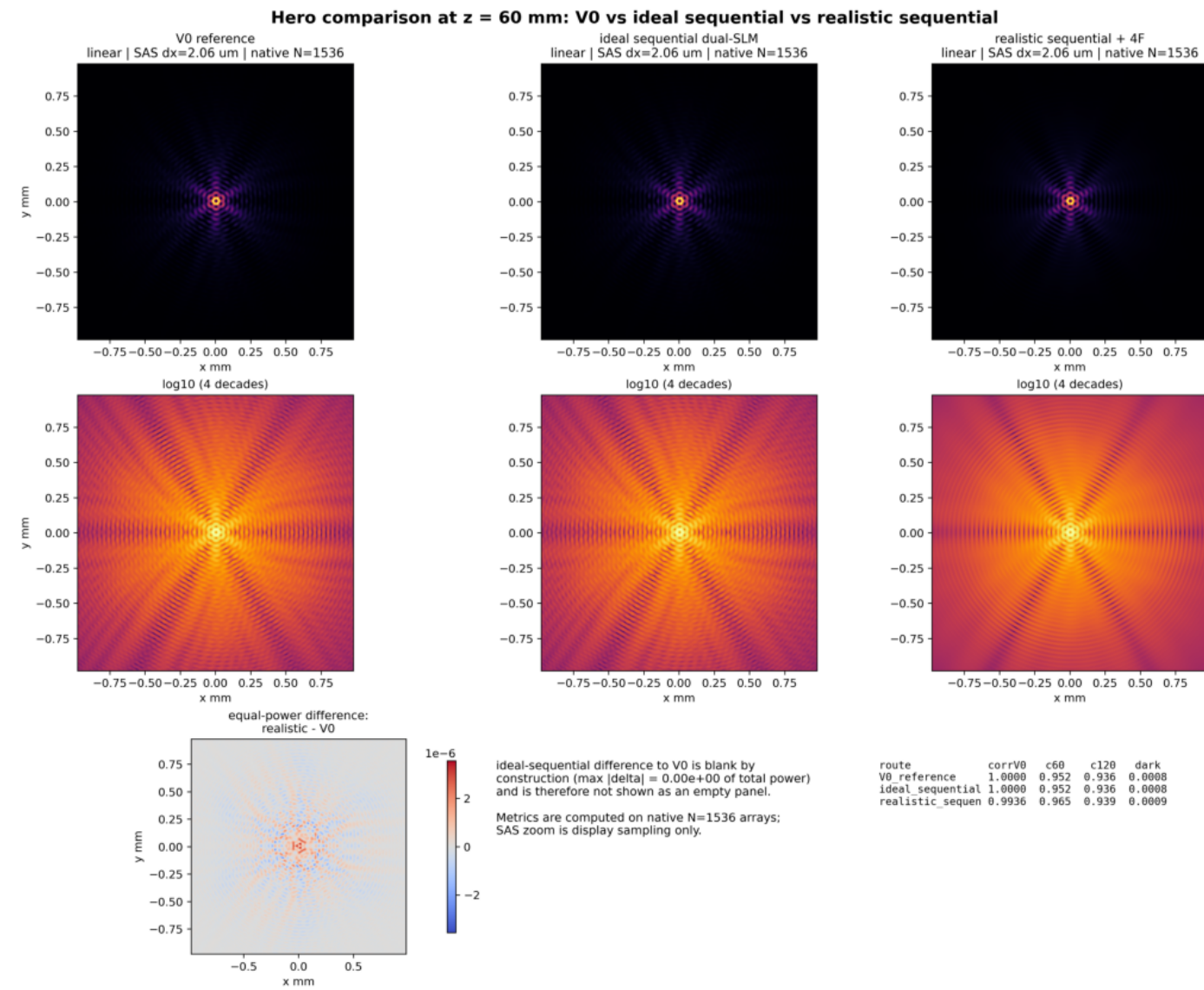
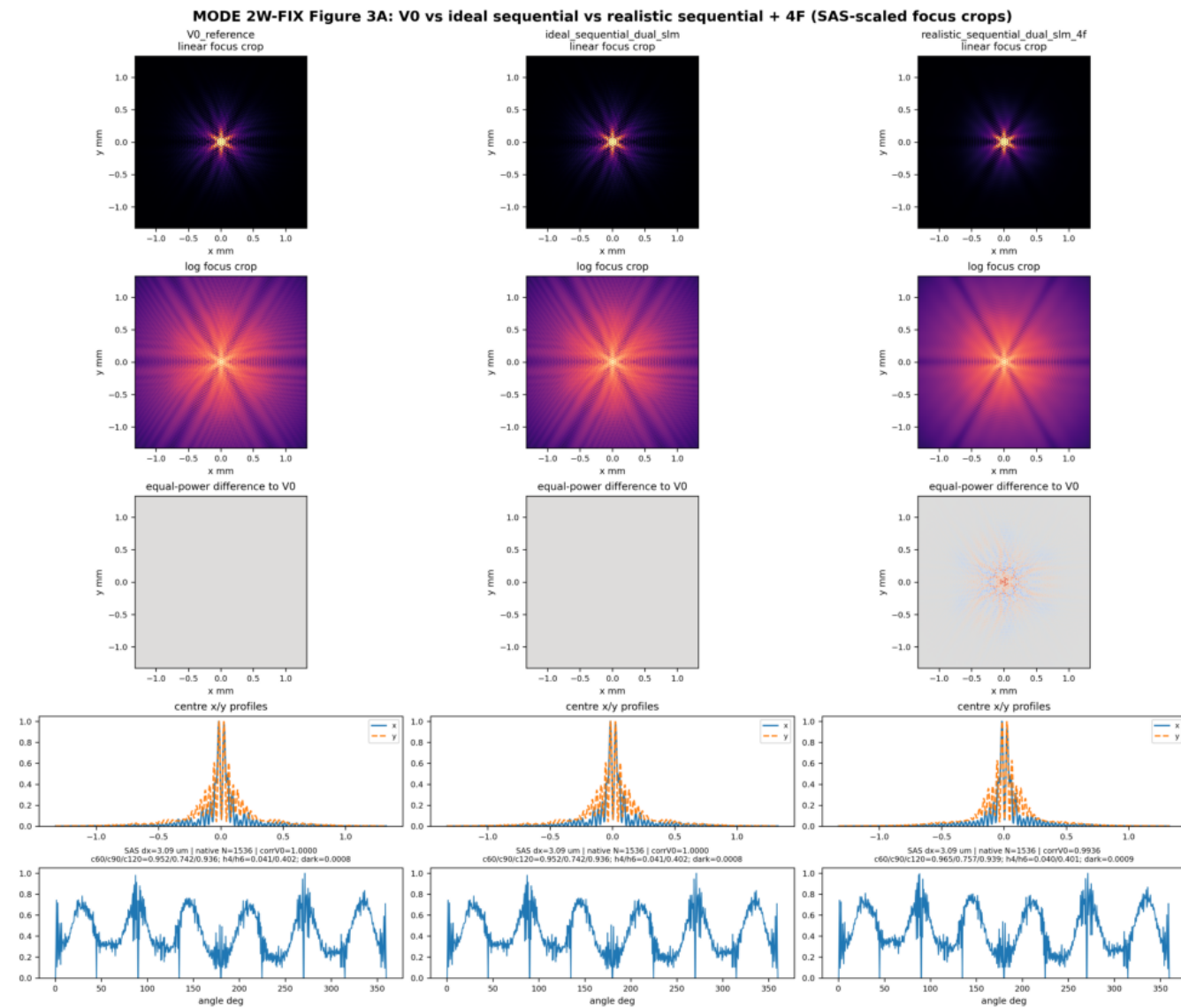
MODE 2W-FIX Figure 2: target vector field and native sequential SLM1/SLM2 masks; LUT not applied



overpacked 5-row hero split into fields (F3A) and quantitative (F3B); blank difference panels replaced by one real panel plus an explicit note

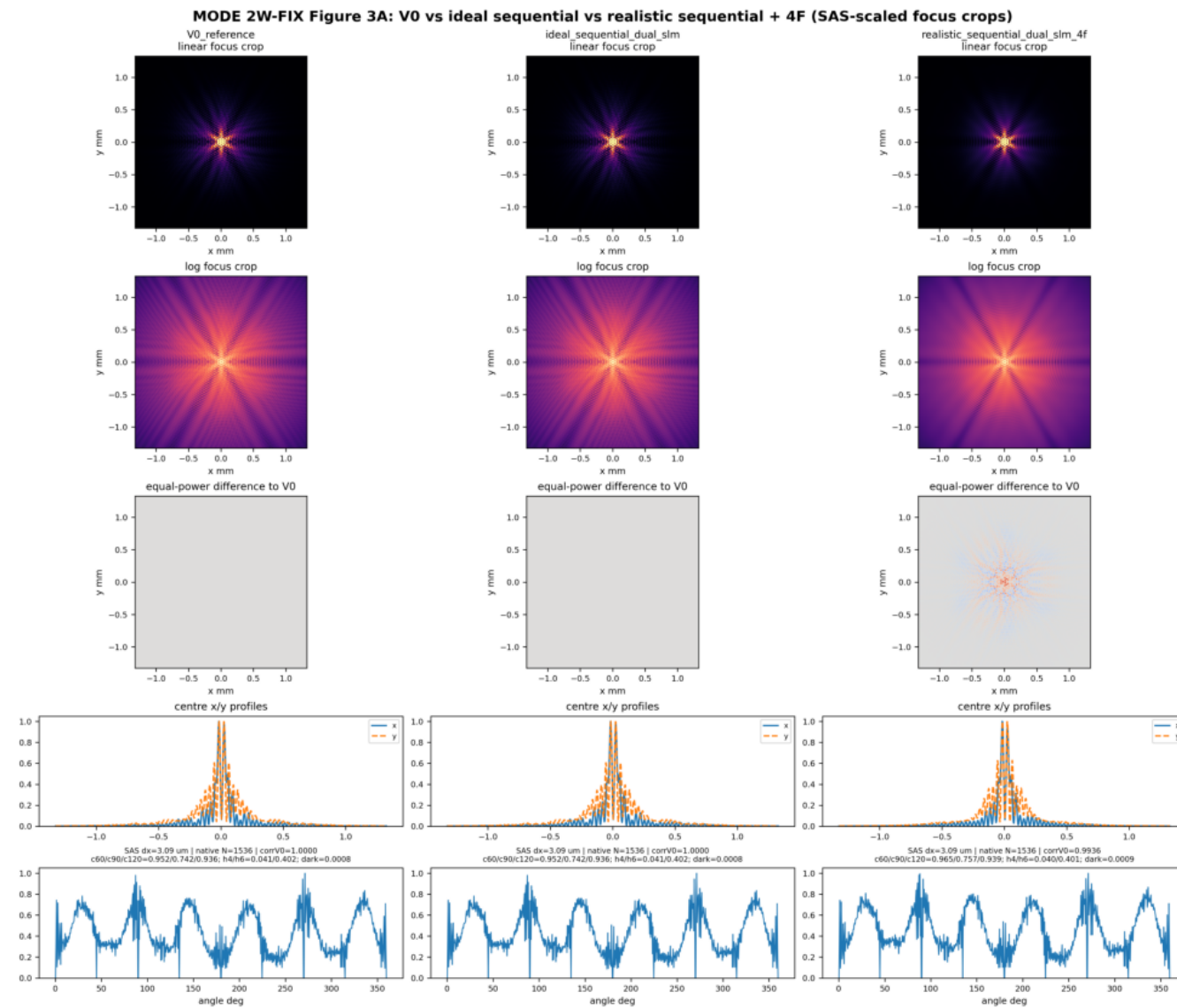
BEFORE: F07

AFTER: F3A_hero_comparison.png



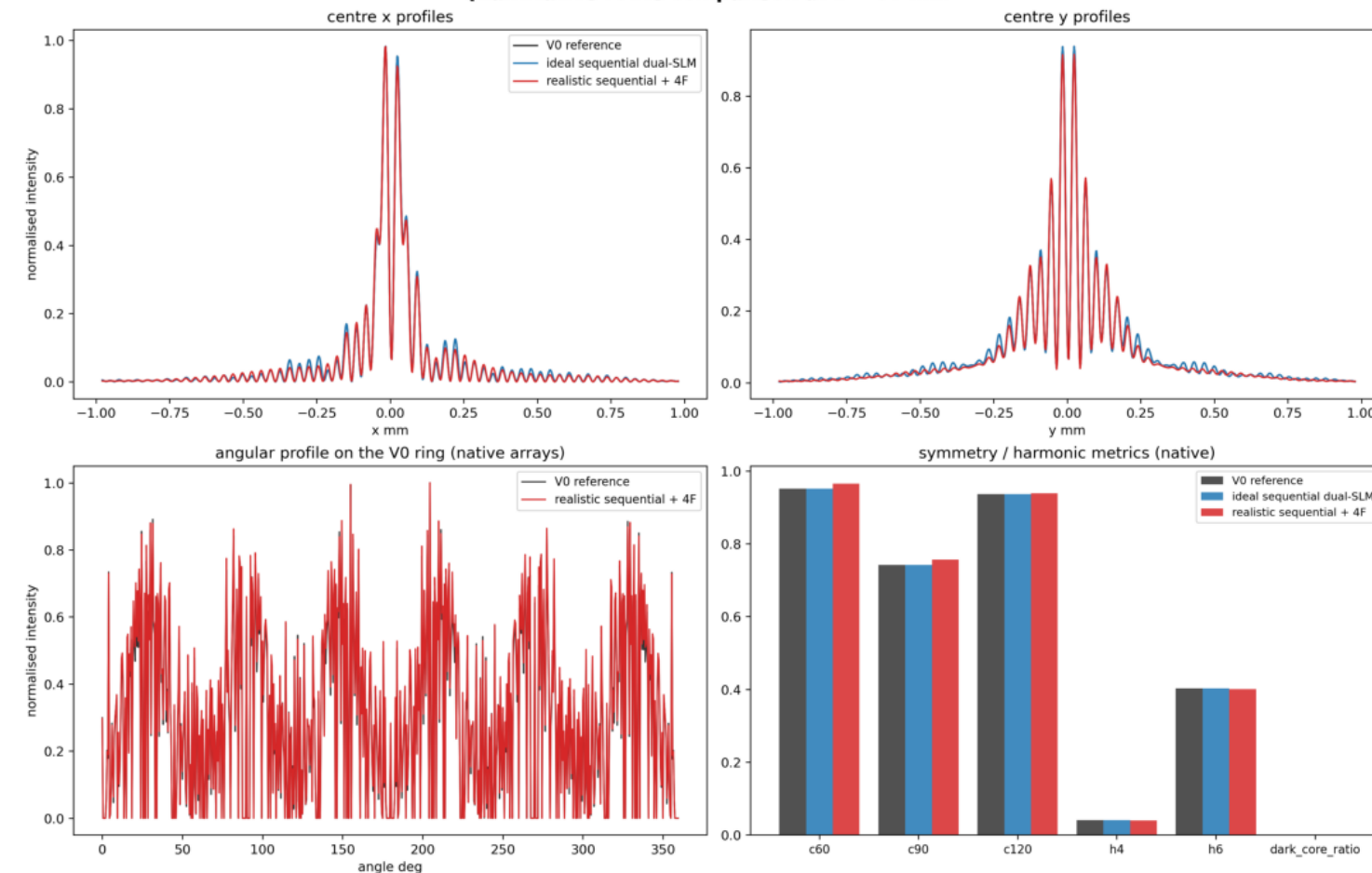
profiles/metrics moved out of the hero into a readable dedicated figure

BEFORE: F07



AFTER: F3B_quantitative_comparison.png

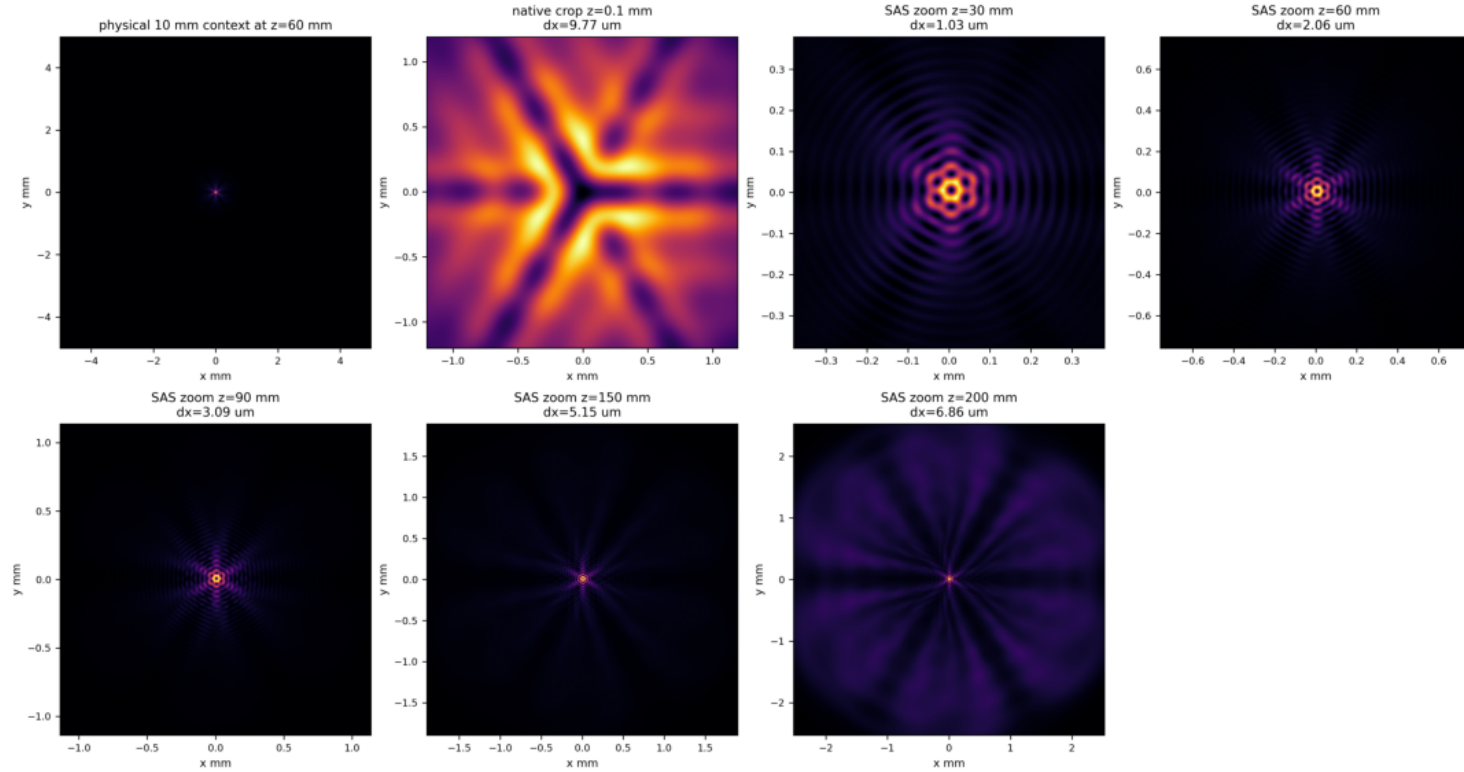
Quantitative route comparison at z = 60 mm



good SAS evolution enlarged with clearer annotations

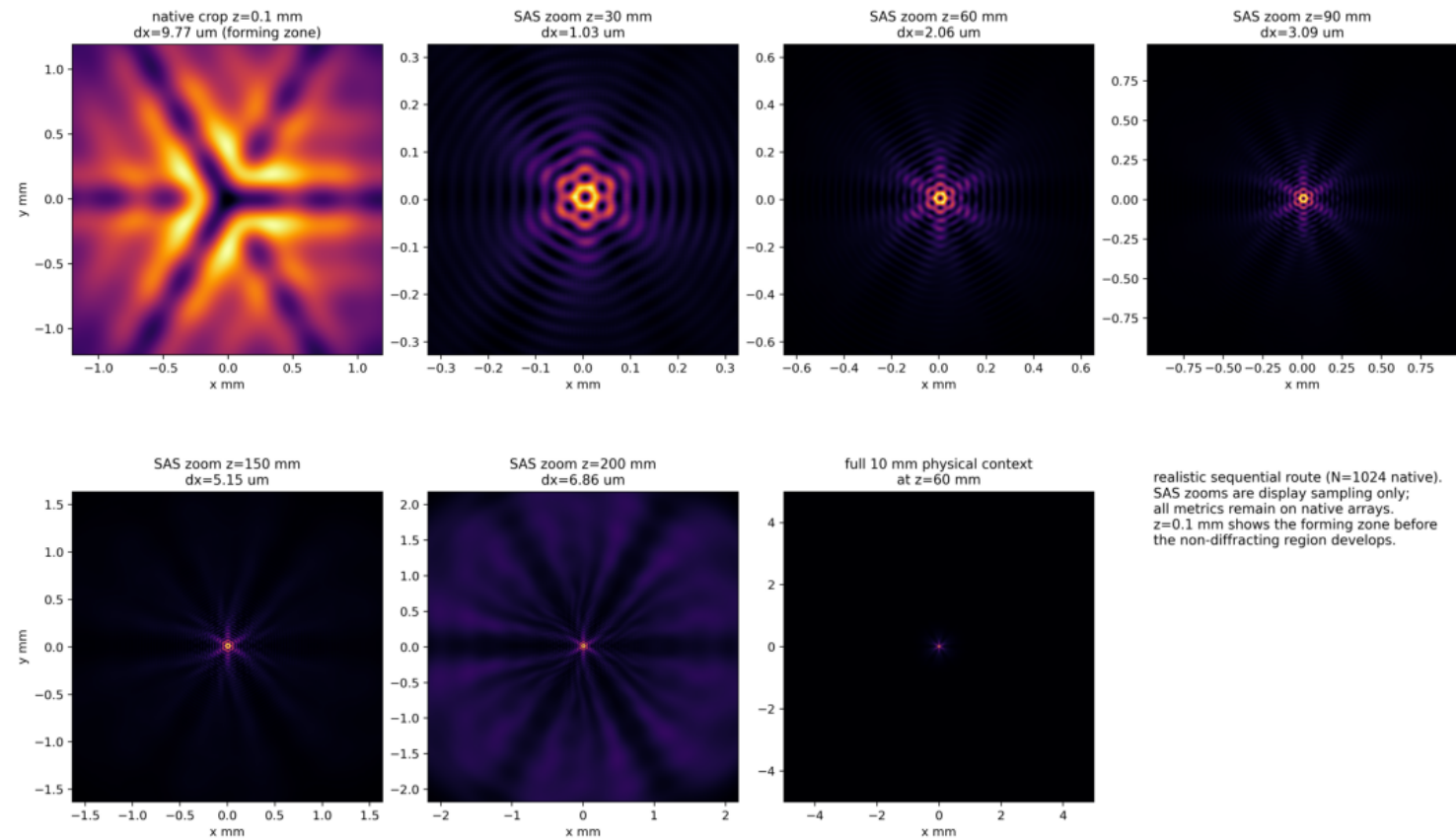
BEFORE: F12

MODE 2W-FIX Figure 4A: transverse evolution with physical context plus SAS-scaled focus crops



AFTER: F4A_transverse_evolution.png

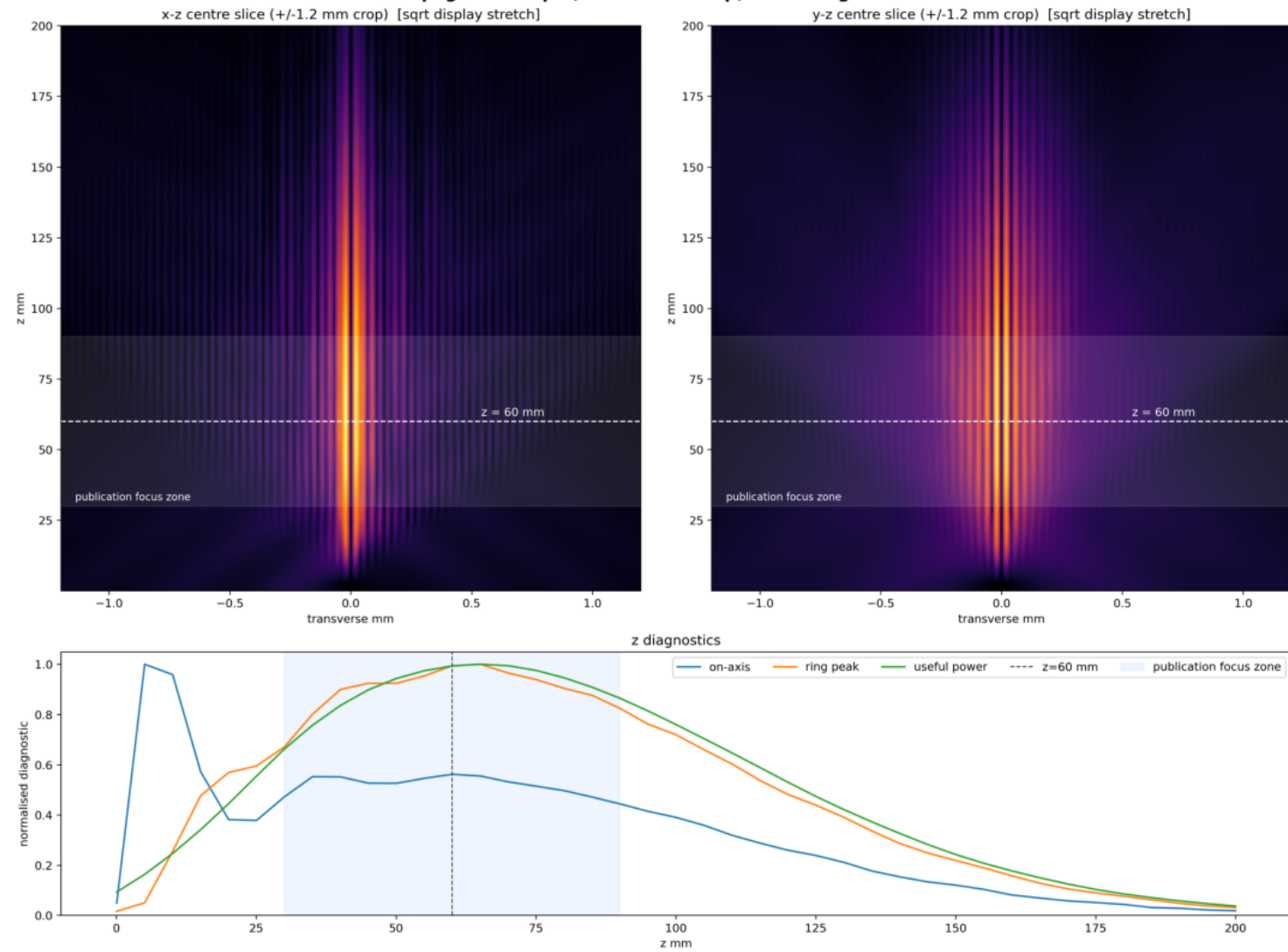
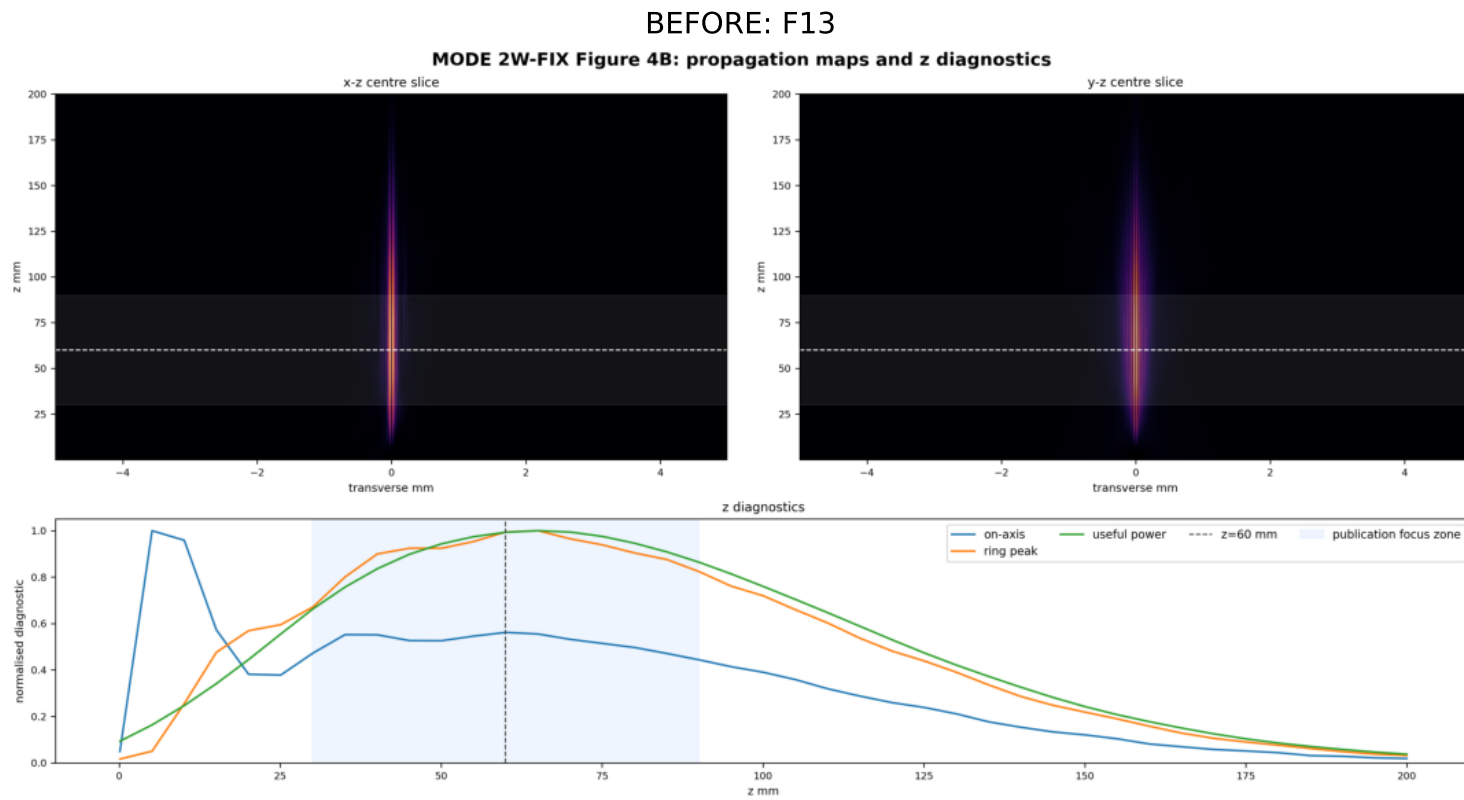
Transverse evolution with physical context (large SAS-scaled focus crops)



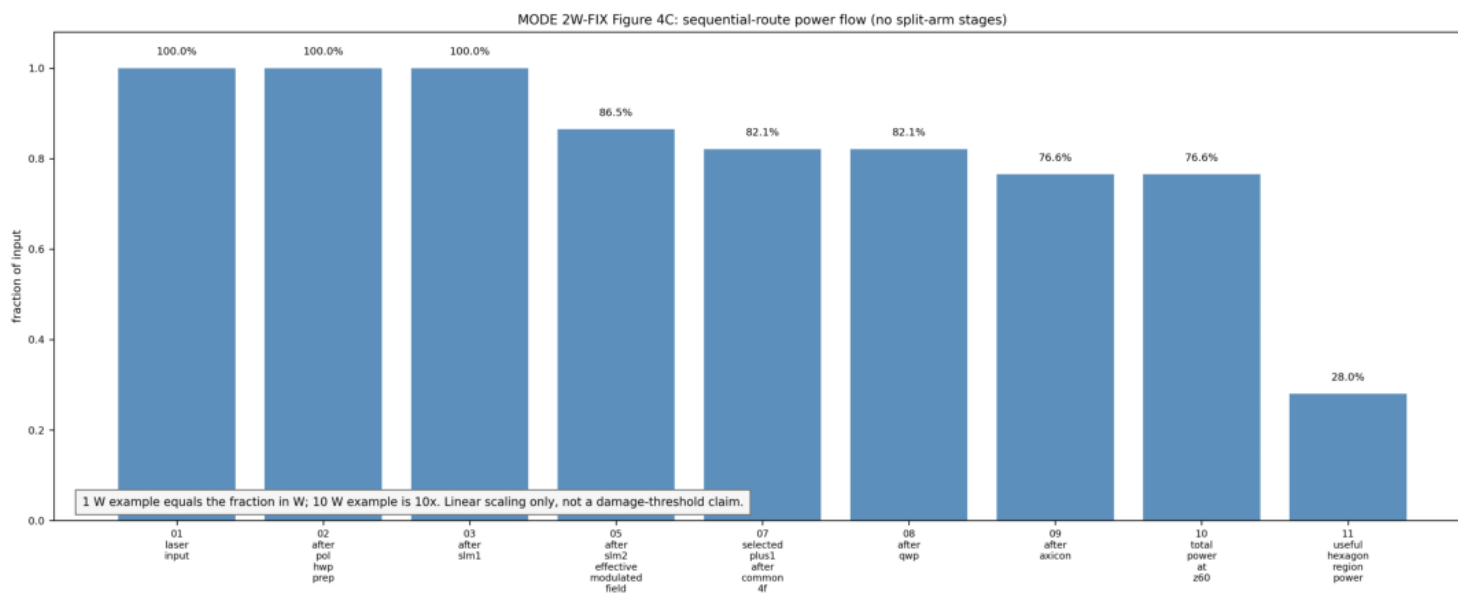
beam occupied ~3% of the +/-5 mm maps; cropped to +/-1.2 mm with sqrt display stretch and zone annotations

AFTER: F4B_xz_yz_propagation.png

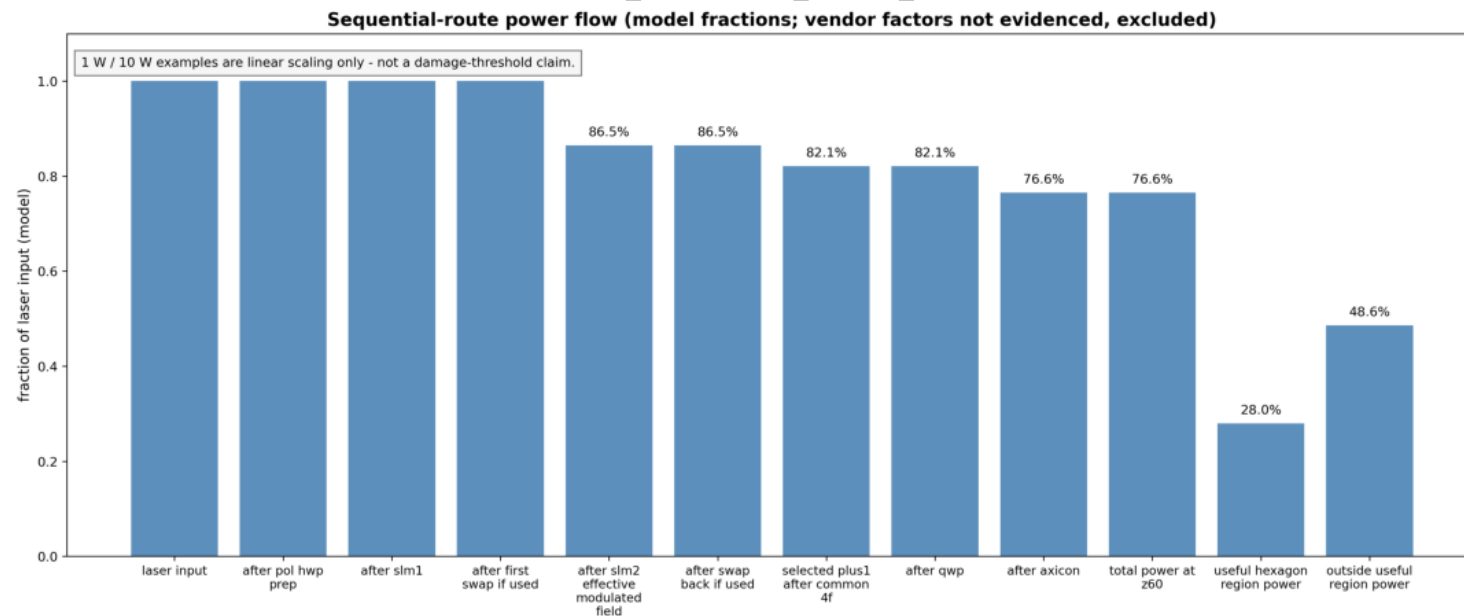
Propagation maps (beam-scale crop) and z diagnostics



BEFORE: F14



AFTER: F4C_sequential_power_flow.png



tiny raster table replaced with a vector table

BEFORE: F15

MODE 2W-FIX Figure 5A: tolerance limits, not pass-fraction bars

parameter	tested range	worst passing	first failure	min corr	strict class
phase quantisation	16 to 1024	1024	not reached	0.9930	visual_hexagonal_field
fill factor	0.93 fixed validated hardware-realism setting	0.93	not reached		visual_hexagonal_field at fixed setting
H/V imbalance	0.8 to 1.2	1.2	not reached	0.9827	visual_hexagonal_field
QWP angle	-2 to 2	-2	not reached	0.9931	visual_hexagonal_field
QWP retardance	-5 to 5	-5	not reached	0.9931	visual_hexagonal_field
iris radius	0.24 to 0.8	0.8	not reached	0.9844	visual_hexagonal_field
iris decentre	-1.5 to 1.5	-1.5	not reached	0.9869	visual_hexagonal_field
H/V registration	0 to 160	160	not reached	0.9753	visual_hexagonal_field
axicon decentre	0 to 1	0.2	0.5	0.7410	triangular_lobed_field, visual_hexagonal_field
z offset	-20 to 20	-20	not reached	0.9791	visual_hexagonal_field

AFTER: F5A_tolerance_limits.png

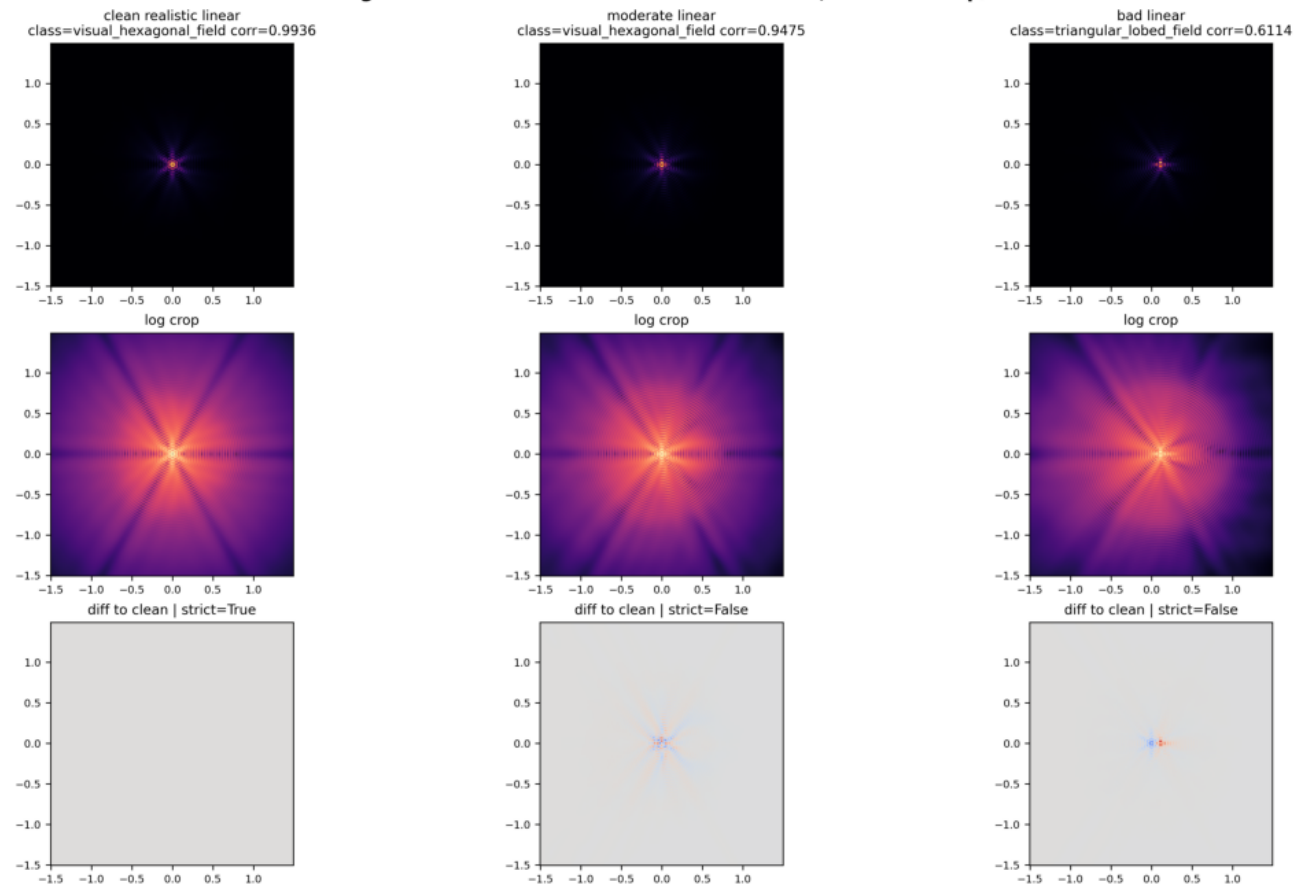
Single-parameter tolerance limits (tested range, worst passing, first failing; strict gate)

parameter	units	tested range	worst passing	first failing	min corr	strict class over sweep
phase quantisation	levels	16 to 1024	1024	not reached	0.9930	visual_hexagonal_field
fill factor	fraction	0.93 fixed validated hardware-realism setting	0.93	not reached		visual_hexagonal_field at fixed setting
H/V imbalance	ratio	0.8 to 1.2	1.2	not reached	0.9827	visual_hexagonal_field
QWP angle	deg	-2 to 2	-2	not reached	0.9931	visual_hexagonal_field
QWP retardance	deg	-5 to 5	-5	not reached	0.9931	visual_hexagonal_field
iris radius	fraction of carrier	0.24 to 0.8	0.8	not reached	0.9844	visual_hexagonal_field
iris decentre	lp/mm	-1.5 to 1.5	-1.5	not reached	0.9869	visual_hexagonal_field
H/V registration	um	0 to 160	160	not reached	0.9753	visual_hexagonal_field
axicon decentre	mm	0 to 1	0.2	0.5	0.7410	triangular_lobed_field, visual_hexagonal_field
z offset	mm	-20 to 20	-20	not reached	0.9791	visual_hexagonal_field

clean/moderate/bad and degraded/corrected merged at identical crop and colour scale

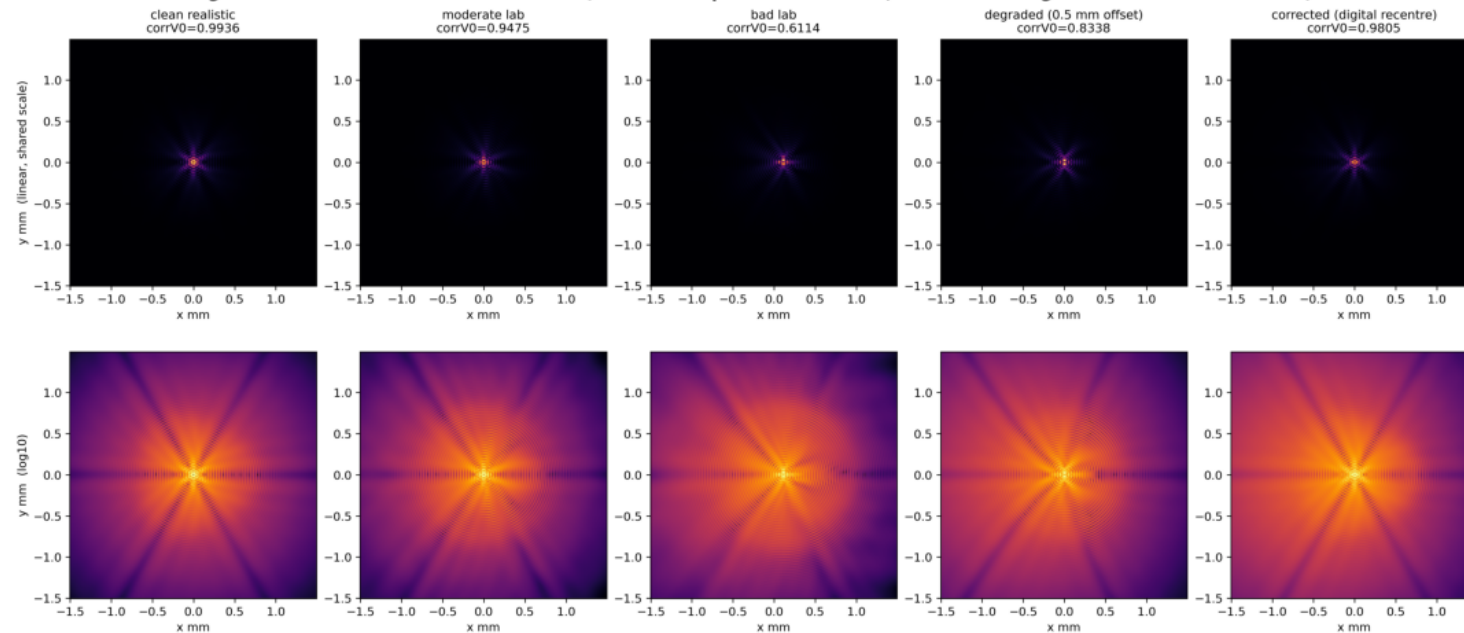
BEFORE: F10

MODE 2W-FIX Figure 3B: clean vs moderate vs bad realism, identical crop/normalisation



AFTER: F5B_degradation_correction.png

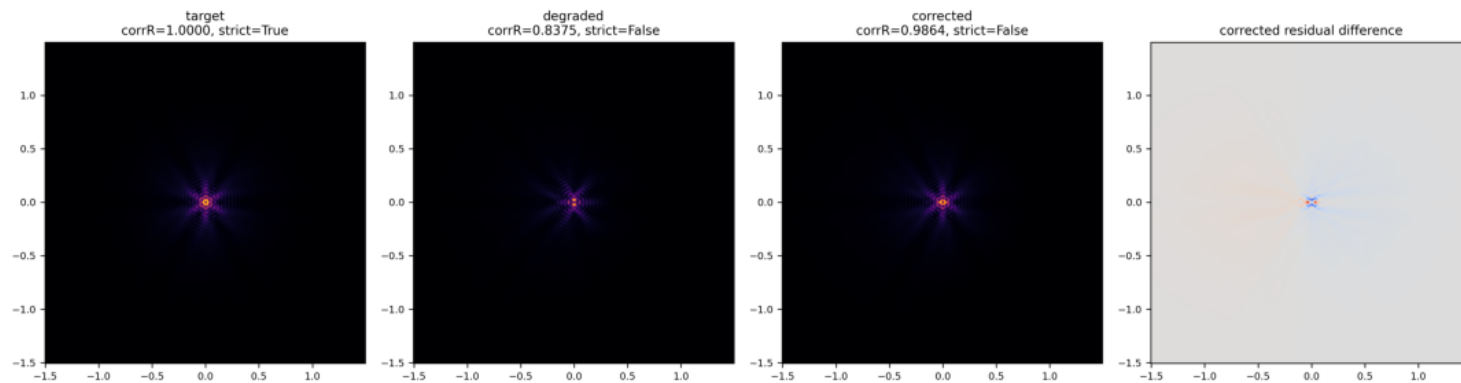
Degradation and correction at z = 60 mm (identical crop and colour scale; correction = digital mask recentre 500 um)



tiny floating panels merged into F5B at shared scale

BEFORE: F11

MODE 2W-FIX Figure 3C: degraded vs corrected recovery



0.5 mm axicon/mask offset: degraded -> corrected by seeded digital recentre. inferred mask_recentre_x=500.0 μm , y=0.0 μm . This is a correction demonstration, not a new physics route.

AFTER: F5B_degradation_correction.png

Degradation and correction at $z = 60$ mm (identical crop and colour scale; correction = digital mask recentre 500 μm)

